

MOUNIKA KATIPALLY

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A recent graduate with a master's degree in Artificial Intelligence, worked on RAG, LLMs, and neural networks. Skilled in machine learning and deep learning frameworks focused on natural language processing. Industry experience in software development and testing has strengthened my problem-solving and engineering skills, adding valuable practical insight. Passionate about applying AI to solve complex problems and eager to contribute to innovative projects while advancing in the field of Artificial Intelligence.

EDUCATION

- University of North Texas | Denton, Texas
 - Graduated: May 2025
 - Master of Science in Artificial Intelligence
 - Minor: Machine Learning
 - Related Coursework: Machine Learning, Large Language Models, Natural Language Processing, Deep Learning, Big Data
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PROJECTS

Analysis and Comparison of RAG Performance of Various LLMs using MedQuAD Data Set

- Developed a Retrieval-Augmented Generation (RAG) pipeline for medical question-answering systems.
- Integrated a hybrid Sentence+T5 retrieval model with re-ranking to improve contextual accuracy and precision.
- Utilized dense vector representations for enhanced understanding and retrieval of medical data.
- Trained and evaluated the system using the MedQuAD dataset.
- Compared performance against large language models using precision, recall, and semantic similarity metrics.
- Achieved improved retrieval efficiency and answer accuracy in healthcare data processing.

Deep Learning for Fault Detection in Semiconductor Manufacturing:

- Developed a Feedforward Neural Network (FNN) for fault detection in semiconductor manufacturing using the UCI SECOM dataset.
 - Implemented Autoencoders and CNNs for feature extraction and high-dimensional data processing.
 - Applied PCA and iterative imputation to handle missing values and improve model accuracy.
 - Evaluated model performance using AUC-ROC, F1-score, precision, and recall optimizing detection accuracy.
 - Compared different models to find the most accurate model for Fault Detection.
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EXPERIENCE

ASSOCIATE SOFTWARE ENGINEER | BENTEC INFO SOLUTIONS PRIVATE LIMITED.

SEPTEMBER 2021 - JULY 2023

- Gathered and analyzed software requirements from internal and external teams.
- Developed and maintained code in Python based on specifications.
- Debugged and upgraded existing software for improved performance and stability.

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- Used GitHub for version control, managed branches for staging, pre-production, and production.
 - Conducted unit testing to validate functionality and detect bugs early.
 - Collaborated with QA to fix bugs and verify features in Agile sprints.
 - Performed automated deployment across environments.
 - Documented changes with clear descriptions and visual comparisons.
 - Supported production releases and ongoing maintenance tasks.

SOFTWARE TESTER | MORADATA QUICK ANALYTICS.

JULY 2019 - JULY 2021

- Experienced in Manual Testing within the IT industry.
- Experienced in creating test plans and test strategies to support manual testing efforts.
- Worked closely with development and support teams for requirement gathering and clarification.
- Involved in collecting, analyzing, and reviewing client requirements to ensure thorough understanding.
- Interpreted Software Requirement Specifications (SRS) to derive comprehensive test scenarios.
- Developed detailed test cases aligned with project requirements and client expectations.
- Prepared test protocols for validating different functional modules.
- Conducted test case reviews to ensure accuracy, completeness, and quality prior to execution.
- Executed functional, smoke, and regression test cases to validate system functionality.
- Identified, logged, and tracked defects/bugs discovered during test execution.
- Utilized JIRA for effective defect reporting, tracking, and management.
- Performed retesting to verify defect fixes and ensure resolution compliance

TECHNICAL SKILLS

- Machine Learning & AI: Neural Networks, Deep Learning, LLMs, Prompt Engineering, Text Classification, Semantic Search, RAG
- Frameworks & Libraries: PyTorch, TensorFlow, Hugging Face, Scikit-Learn, OpenAI APIs
- Data Visualization Tools: Tableau, PowerBI, Matplotlib
- Code Assistants: Cursor, GitHub Copilot
- Programming: Python, MATLAB, SQL, Java, C
- Data & Evaluation: Model Tuning (AUC-ROC, F1), Data Imputation, PCA, Pandas, NumPy, Jupyter Notebook
- Retrieval & Databases: FAISS, BM25, Embedding-based Retrieval, SQL
- Cloud: AWS (S3, EC2), Apache Spark
- Testing & QA: Manual Testing, Regression & Smoke Testing, Bug Tracking, Test Documentation
- Collaboration Tools: Git, GitHub, Postman, JIRA, VS Code

